

Statistical Distributions and Combinatorial Optimization for Submillimetric MR-CT-STL Craniofacial Registration

Cartik Sharma, Steve Mann, Senior Member, IEEE, and Alexander Vicol

Department of Electrical & Computer Engineering, University of Toronto, Toronto, ON, M5S 3G8, Canada

Emails: cartik.sharma@gmail.com, mann@eyetap.org, alexander.vicol@mail.utoronto.ca

Abstract—

Precise submillimetric alignment across magnetic resonance imaging (MRI), computed tomography (CT), and physical stereolithography (STL) meshes is crucial for stereotactic neurosurgery, guided cranial repair, and brain-computer interface (BCI) cap fitting. Classical registration techniques, such as the Iterative Closest Point (ICP) algorithm, are highly susceptible to local minima traps and perform poorly under significant cross-modality noise and geometric scaling disparities. This paper presents a mathematically robust registration framework integrating continuous statistical density estimations with discrete combinatorial optimization. We formulate the spatial distributions of point clouds using multivariate Gaussian, Laplace, and Student's *t* density models. Node-to-node correspondence is subsequently resolved as a weighted bipartite graph matching problem solved via the Hungarian combinatorial assignment algorithm. Sizing mismatches are corrected using a continuous-state differential continued-fraction transform scaler. Empirical validation on clinical cranial marching cubes reconstructions and laser-scanned surgical STL models proves that GMM statistical fusion achieves a Target Registration Error (TRE) of 0.142 mm, while Student's *t* and Laplace distributions coupled with Hungarian matching achieve submillimetric errors of 0.068 mm and 0.089 mm respectively. Statistical risk measures, including Value at Risk (VaR 95%) and Conditional Value at Risk (CVaR 95%), demonstrate that the combinatorial matching pipeline exhibits exceptional resilience against outliers, ensuring safe, submillimetric medical device fusion.

Index Terms—Bipartite matching, combinatorics, cranial surgery, continued fractions, Gaussian mixture models, medical image registration, MRI-to-CT, stereolithography (STL), Target Registration Error (TRE), Value at Risk.

I. Introduction

COMPUTATIONAL craniofacial navigation and stereotactic guidance systems rely on accurate alignment between anatomical volumes and physical structures. Magnetic Resonance Imaging (MRI) is standard for visualizing soft-tissue boundaries, whereas Computed Tomography (CT) captures high-density osseous details. Physical surgical guides, implants, and wearable EEG arrays are represented as stereolithography (STL) meshes. Fusing these three modalities—MR, CT, and STL—into a singular, unified coordinate system is essential for stereotactic interventions.

Traditional registration routines rely heavily on the Iterative Closest Point (ICP) algorithm or its variants. ICP operates by establishing hard, nearest-neighbor matching and solving a least-squares rigid transform. However, when registering downsampled MRI cortical models to high-density laser-scanned surgical STL meshes, ICP is prone to local minima traps and convergence failure. This is caused by missing structural details, noise, and initial orientation scaling disparities, yielding Target Registration Errors (TRE) often exceeding 5 mm.

To address these issues, we propose a hybrid framework combining statistical distributions with discrete combinatorial optimization. Instead of hard nearest-neighbor pairs, we define probabilistic density fields over the manifolds. We construct a bipartite graph matching formulation to resolve point correspondence globally rather than locally, and we use rational continued fraction convergents to approximate scaling and rotation

matrices with exact rational numbers. This paper presents the finite mathematical equations governing these paradigms and details their submillimetric empirical performance.

II. Statistical Distribution Formulation

A. Multivariate Gaussian Mixture Models

Let the source point set (e.g. reconstructed MRI cortex marching cubes) be denoted as $X = \{x_1, \dots, x_N\}$ where x_i in $\mathbb{R}^3$, and the target set (e.g. CT or STL mesh vertices) as $Y = \{y_1, \dots, y_M\}$ where y_j in $\mathbb{R}^3$. We represent X as the centroids of a Gaussian Mixture Model (GMM). The probability density function is expressed as:

$$p(x) = \sum_{k=1}^K \pi_k \tilde{N}(x \mid \mu_k, \Sigma_k)$$

where π_k are mixing coefficients satisfying $\sum_k \pi_k = 1$, and $\tilde{N}(x \mid \mu_k, \Sigma_k)$ represents the trivariate Gaussian component. To solve for rotation R and translation t , we maximize the log-likelihood of the observed target dataset Y under the probability distribution, which minimizes the Kullback-Leibler (KL) divergence. In the Expectation-Maximization (EM) algorithm, the posterior probabilities (membership weights) p_{ik} are computed in the E-step:

$$p_{ik} = \pi_k \tilde{N}(y_i \mid R\mu_k + t, \Sigma_k) / \left[\sum_{j=1}^K \pi_j \tilde{N}(y_i \mid R\mu_j + t, \Sigma_j) \right]$$

B. Laplace and Student's *t* Distributions

Standard GMMs are highly sensitive to extreme outliers, such as boundary scanning artifacts. To resolve this, we replace the Gaussian kernel with heavy-tailed multivariate Laplace or Student's *t*-distributions. The probability density function for a Student's *t*-distribution with ν degrees of freedom is formulated as:

$$t(x \mid \mu, \Sigma, \nu) = \frac{[\Gamma((\nu+3)/2) / (\Gamma(\nu/2)(\nu\pi)^{3/2}|\Sigma|^{1/2})]}{[1 + (1/\nu)(x-\mu)^T \Sigma^{-1}(x-\mu)]^{-(\nu+3)/2}} \cdot$$

where $\Gamma(\cdot)$ is the gamma function. When $\nu \rightarrow 1$, the distribution converges to a Cauchy kernel, providing maximum robustness against outlying surface segments. The parameter ν is optimized dynamically.

III. Combinatorial Bipartite Matching

While statistical distributions capture global density overlays, they do not guarantee unique topological point correspondence. This can lead to local mesh folding or structural collapse. We solve this by modeling point alignment discretely as a maximum-weight bipartite graph matching problem. Let $G = (X \cup Y, E)$ be a bipartite graph where edges connect source points X to target points Y .

We define a combinatorial cost matrix C in $\mathbb{R}^{(N \times M)}$, where each entry c_{ij} represents the penalty of aligning node x_i to node y_j . To capture both spatial proximity and structural topology, c_{ij} combines Euclidean distance with the KL divergence of localized statistical density functions D_i and D_j :

$$c_{ij} = ||x_i - y_j||_2^2 + \gamma \cdot \text{KL}(D_i || D_j)$$

where $\gamma \geq 0$ is a hyperparameter control. The optimal assignment is represented by a binary matrix X_{mat} where $x_{ij} = 1$ if node x_i maps to y_j , and 0 otherwise. We solve for the matching that minimizes total bipartite cost:

$$\text{Minimize } \sum_{i=1}^N \sum_{j=1}^M c_{ij} x_{ij}$$

subject to the matching constraints:

$$\sum_{j=1}^M x_{ij} \leq 1, \forall i; \quad \sum_{i=1}^N x_{ij} \leq 1, \forall j; \quad x_{ij} \in \{0, 1\}$$

This optimization is solved in polynomial time $O(V^3)$ using the Hungarian algorithm, which constructs a set of alternating paths over the bipartite layout to find the exact global minimum. This guarantees that each cortical segment corresponds to a unique target implant zone, preventing topological collapse.

IV. Continued Fraction Transformation

To achieve submillimetric precision, floating-point rounding errors must be eliminated. Rigid coordinate transformation contains scaling factor s and rotation matrix R . We represent the scaling ratio s as a rational number approximated using finite continued fraction expansion. A continued fraction represents a real number via a sequence of integer convergents $[a_0; a_1, a_2, \dots, a_p]$:

$$s \approx a_0 + 1 / (a_1 + 1 / (a_2 + 1 / (a_3 + \dots + 1/a_p)))$$

The rational convergents p_n / q_n are evaluated iteratively using the recurrence relations: $p_n = a_n p_{n-1} + p_{n-2}$ and $q_n = a_n q_{n-1} + q_{n-2}$, starting from $p_{-1} = 1, p_{-2} = 0, q_{-1} = 0$, and $q_{-2} = 1$. By truncating the expansion at convergent index p , we establish scaling operations with exact rational values, avoiding cumulative precision loss across large meshes.

V. Empirical Results & Discussion

We implemented the statistical and combinatorial matching pipelines in Python within the Mersivity suite. We registered clinical T1-weighted MRI cortical reconstructions and high-resolution Computed Tomography (CT) voxel stacks to physical titanium cranial implant STL meshes. The source and target meshes consisted of $N = M = 2048$ sampled nodes.

Table I details the quantitative results. The initial Target Registration Error (TRE) prior to registration was 21.30 mm. GMM statistical registration ($K = 6$ components) achieved a TRE of 0.142 mm in 1.25 seconds. Laplace distribution alignment combined with Hungarian bipartite matching reduced the TRE to 0.089 mm in 1.84 seconds. The Student's t-distribution ($v = 1.8$) coupled with Hungarian matching achieved the highest submillimetric accuracy, yielding a final TRE of **0.068 mm**.

Distribution	Matching	Final TRE	VaR 95%	CVaR 95%
Gaussian	None (EM)	0.142 mm	0.245 mm	0.312 mm
Laplace	Hungarian	0.089 mm	0.134 mm	0.178 mm
Student's t	Hungarian	0.068 mm	0.098 mm	0.124 mm

Table I: Comparative Target Registration Error (TRE) and risk metrics across statistical and matching methods.

To evaluate registration safety under clinical noise, we calculated Value at Risk (VaR) and Conditional Value at Risk (CVaR) at a 95% confidence level. Let the loss L be the distribution of point-to-plane residual errors. $\text{VaR}_{0.95}$ represents the threshold below which 95% of registration errors fall. $\text{CVaR}_{0.95}$ computes the mean of the worst 5% of registration errors:

$$\text{VaR}_{\alpha}(L) = \inf \{ l \in \mathbb{R} \mid P(L > l) \leq 1 - \alpha \}$$

$$\text{CVaR}_{\alpha}(L) = E[L \mid L \geq \text{VaR}_{\alpha}(L)] = (1/(1-\alpha)) \int_{\alpha}^1 \text{VaR}_u(L) du$$

As shown in Table I, GMM alignment yielded a CVaR of 0.312 mm, suggesting that outlier regions experience larger mismatch. Conversely, the Student's t distribution with Hungarian matching restricted the worst-case CVaR to just 0.124 mm. This indicates a highly stable, uniform deformation across all craniofacial regions, securing the submillimetric margins required for robotic bone cutting and cortical cap placement.

VI. Conclusion

This paper has presented a hybrid framework integrating statistical density modeling with combinatorial optimization for multi-modal MR-CT-STL craniofacial registration. By utilizing Student's t distributions, we achieved robustness against outlier scanning noise. Solving node correspondence globally using the Hungarian algorithm prevents local mesh folds. Furthermore, representing scale transforms via rational continued fraction convergents eliminates numerical precision loss. Our pipeline converges to a final Target Registration Error (TRE) of 0.068 mm with a worst-case CVaR of 0.124 mm. This satisfies the submillimetric constraints of guided surgery, and future work will focus on embedding these solvers into active real-time surgical robotics.

References

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